Ocean shipping relies a lot on fossil fuels, and it's one of the hardest areas to decarbonize. Green methanol can be a possible solution. It's methanol made by reacting captured carbon dioxide with green hydrogen, where the hydrogen is produced by water electrolysis powered by low carbon electricity. Methanol is useful because it's already traded globally, and it can be used as a fuel.

Here, we're trying to find out whether green methanol can scale in a way that's both good for climate and that's economically realistic. The chemistry can look straightforward but the full process really isn't. Carbon dioxide has to be captured, purified, and compressed. Hydrogen production uses a lot of electricity, and the emissions benefit depends a lot on the electricity source. The methanol synthesis reactor must handle heat release, manage recycle streams, and achieve high conversion without too much energy use. Then, product purification needs a lot of energy.

Chemical engineers can contribute a lot by modeling the process from end to end and by finding out which steps matter most for cost and emissions. we can compare carbon dioxide sources like cement and steel flue gas, biogenic streams, and we can test how location and purity makes this more or less realistic. We can estimate mass and energy balances, then connect them to rough costs and to constraints like water supply for electrolysis.

Project Questions

what's the most realistic market for green methanol, and what changes would be needed for adoption to grow?

which process step would be the best when using realistic assumptions?

what are the economic, societal, environmental, and cultural tradeoffs?

what's the most likely failure mode for this idea?

Process flow diagram

CO_2 source -> capture and purification -> compression -> water electrolysis (H_2 + O_2) -> methanol synthesis reactor with recycle -> condensation -> distillation -> methanol storage and transport

Calculation

Goal: estimate material needs and energy and electricity cost.

Overall reaction:

$CO_2 + 3H_2 \rightarrow CH_3OH + H_2O$

Molar masses (g per mol):

CO_2 = 44

H_2 = 2

CH_3OH = 32

H_2O = 18

For 1,000 kg methanol:

Moles of methanol = 1,000 kg / 32 kg per kmol = 31.25 kmol

CO_2 needed = 31.25 kmol * 44 kg kmol = 1,375 kg CO_2

H_2 needed = 3 * 31.25 kmol * 2 kg per kmol = 187.5 kg H_2

water produced = 31.25 kmol * 18 kg per kmol = 562.5 kg H_2O

inputs:

1.375 metric tons CO_2

187.5 kg H_2

2) **Electrolysis electricity for the hydrogen**

Assumption: 55 kwh per kg H_2.

Electricity = 187.5 kg * 55 kwh per kg = 10,312.5 kwh

that’s 10.3125 Mwh per ton methanol

Electricity cost for electrolysis only:

at \$0.03 per kwh: 10,312.5 * 0.03 = \$309.38

at \$0.05 per kwh: 10,312.5 * 0.05 = \$515.63

at \$0.10 per kwh: 10,312.5 * 0.10 = \$1,031.25

3) water use and oxygen coproduct from electrolysis

minimum water consumption comes from: $2H_2O \rightarrow 2H_2 + O_2$
mass ratio water to hydrogen is around 18 to 2, which is 9 to 1

water needed = 9 kg water per kg H_2 * 187.5 kg H_2 = 1,687.5 kg water

That's about 1.69 cubic meters of water minnimum

Oxygen produced uses an 8 to 1 mass ratio (O_2 to H_2).

O_2 produced = 8 * 187.5 kg = 1,500 kg O_2

4) **Energy**

Assumptions:

CO_2 capture and purification energy: 2.0 GJ per ton CO_2

CO_2 compression: 100 kwh per ton CO_2

Methanol separation and distillation: 1.0 GJ per ton methanol

conversion: 1 GJ = 277.78 kwh

capture energy = 2.0 GJ per ton CO_2 * 277.78 kwh per GJ * 1.375 tons CO_2
= 763.89 kwh

compression energy = 100 kwh per ton CO_2 * 1.375 = 137.5 kwh

distillation energy = 1.0 GJ per ton methanol * 277.78 = 277.78 kwh

total energy:

total = 10,312.5 + 763.89 + 137.5 + 277.78

total = 11,491.67 kwh, which is 11.49 Mwh per ton methanol if you price all of that as electricity:

at $0.05 per kwh: 11,491.67 * 0.05 = $574.58 per ton methanol

if a plant makes 1,000 tons methanol per day, the electrolysis electricity alone is around 10.3 Mwh per ton * 1,000 per day = 10,300 Mwh per day, which averages to about 430 mw

Tradeoffs:
The economics depend on electricity price, electrolyzer cost, and how far CO2 must be transported. The societal side is mostly about issues like truck traffic, safety zoning, and who gets jobs versus who takes on risk. Environmental tradeoffs are like water use, local air quality near ports, and whether electricity is actually low carbon. Cultural and public acceptance also matters a lot, because new industrial facilities near ports can face public issues even if the climate case is strong.

Part 3

1) Prompt used in LibreChat

Topic: producing methanol from captured carbon dio*ide and green hydrogen made by water electrolysis.

Please produce:

An introduction of 250 to 300 words that e*plains why the topic matters and what chemical engineers do in this space.

Two to four project questions. At least one must explicitly require either a basic calculation or a process flow diagram.

A calculation section that uses $CO_2 + 3H_2 \rightarrow CH_3OH + H_2O$ to compute CO_2 and H_2 required for 1 metric ton of methanol. Assume 55 kWh per kg H_2 for electrolysis and compute electricity cost at \$0.03, \$0.05, and \$0.10 per kWh. State assumptions.

2) GenAI response:

Introduction

Green methanol may be evaluated as a coupled process system rather than as an isolated reaction—and the primary difficulty is not the governing equation—it is the specification of the system boundary, the data boundary, and the control boundary. If these boundaries are not defined a priori—then the "green" designation becomes non-stationary.

A chemical engineering framing renders the problem tractable—by enforcing explicit assumptions and explicit interfaces between unit operations. Inlet purity and pressure targets are specified—utilities and their availability are specified—control variables and acceptable operating ranges are specified—and response behavior under deviation is specified. The design is then stress-tested against representative disturbances—variable electricity pricing and availability, CO2 supply fluctuations, non-ideal separations, downtime and ramp-rate limits, and safety and permitting requirements that constrain layout and operations. The deliverable is not an optimal or "perfect" design—it is a compact process map with declared assumptions—a baseline mass and power estimate establishing a lower bound—and a prioritized set of missing measurements required before the concept may be treated as scalable rather than hypothetical.

Project questions

1. What boundary rules can be used for lifecycle accounting and verification—what gets counted—what gets excluded—and how can you audit it?

2. What operating mode do you assume—steady operation, load following, or scheduled runs—and what control variables keep product quality stable?

3. What hazards dominate the plant design—what is the top HAZOP concern—and what design feature addresses it?

Calculation

Reaction: $CO_2 + 3H_2 \rightarrow CH_3OH + H_2O$
Basis: 1,000 kg CH_3OH

- CO_2 needed = (44/32)×1,000 = 1,375 kg
- H_2 needed = (6/32)×1,000 = 187.5 kg
  Assume electrolysis = 55 kWh per kg H_2
- Electricity = 187.5×55 = 10,312.5 kWh

Electricity cost:

- \$0.03 per kWh → \$309.38
- \$0.05 per kWh → \$515.63
- \$0.10 per kWh → \$1,031.25

Extra scale check—power requirement at 1,000 tons per day:

- Daily electricity = 10.3125 MWh/ton × 1,000 ton/day = 10,312.5 MWh/day
- Average power = 10,312.5 ÷ 24 = 429.7 MW

My evaluation of the AI answer:

The AI answer is nice because it frames green methanol as a systems design and verification problem, and its questions focus on lifecycle boundaries. The stoichiometry and electricity cost math is correct for a minimum case, and the 1,000 tons per day scale check helps show how large the power demand could be. The intro is a little short, and the calculation leaves out energy uses like CO_2 capture, compression, recycle compression, and distillation.